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Evolution and economic complexity: an overview

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Evolution and economic complexity: an overview

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The articles in this special issue are based upon a selection of papers presented at the 'Brisbane Club' section of the 2007 STOREP Conference. We provide introductory reviews of these articles and briefly assess their contributions to the advancement of our understanding of economic evolution, and the associated process of innovation. In particular, we discuss why it is necessary to adopt a complex systems perspective, which deals explicitly with the creation and transmission of new knowledge, as the analytical foundation of evolutionary economic modelling.

Keywords: agent-based modelling; complexity; complex system; evolutionary economics; innovation; knowledge; networks

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In 1999, the first meeting of the Brisbane Club was held at the University of Queensland in Brisbane, hosted by both of us. The intention of the workshop was to open up discussion and debate concerning the links and possible points of development and integration between neo-Schumpeterian evolutionary economics and emerging complex systems approaches to economic analysis. This meeting was highly successful in terms of setting an agenda for work at the interface of economic evolution and complexity and gave rise to the Foster and Metcalfe (2001) volume. Further meetings were held in: Manchester in 2002, yielding the Metcalfe and Foster (2004) volume; Brisbane in 2005; Pollenzo in 2007, in conjunction with the annual STOREP conference. The articles in this special issue are revised versions of papers presented at Pollenzo. These articles remain firmly in the Brisbane Club tradition, each seeking to provide complex system insights to augment and deepen evolutionary economics. We are particularly grateful to Cristiano Antonelli, and the facilitators of the STOREP meeting for making this event possible.

In the first article, Cristiano Antonelli provides an engrossing examination of how modern complexity approaches developed from earlier work on the economics of innovation. He provides an overview of research on the economics of innovation up to the present day when complex systems thinking is pervading the field in fundamental ways.

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He identifies four heuristic frameworks stemming from: the 'classical' contributions of both Adam Smith and Karl Marx, whereby new technologies are intentionally introduced in response to 'demand-pull'; from Joseph Schumpeter, who connected innovation and competition; from Kenneth Arrow, who connected innovation with knowledge; from the evolutionary approaches that began with biological analogies and, ultimately, became connected with modern complex systems thinking. A key contribution is made in pointing out that, although there is a clear focus in each case, there are many connections between these four strands of thinking and that these have yielded an integrated evolutionary approach to the economics of innovation. The notion of 'path dependence', which can be discerned in various guises in all of the strands, is identified as providing a key conceptual tool that can be used to engage in coherent analysis of the behaviour of complex economic systems. Path dependence is viewed as reflecting the basic laws that operate in complex systems and, thus, is seen as a basic building block in understanding the process of economic growth.

It is commonplace in the literature on economic growth today to assert the importance of knowledge. As is pointed out by Foster (2005), in third- and fourth-order complex systems, the acquisition and sharing of knowledge is what distinguishes these from second-order complex systems. Thus, as Urich Witt and Christian Zellner point out in their article, this means the discovery and transmission of knowledge are fundamentally important processes that are generally ignored in the conventional endogenous growth literature where simplistic analytical representations of economic systems are relied upon. Both the discovery and transmission of knowledge in a complex system involve entrepreneurship. Indeed, much of entrepreneurship is concerned with finding ways of connecting up fragments of knowledge to produce a commercially viable innovation. The distinction between 'procedural' and 'tacit' knowledge turns out to be crucial to the way that knowledge is discovered, transmitted and applied. Witt and Zellner contrast the quite different cases of the transfer of knowledge in start-up companies, by former scientists and engineers, compared with that which occurs in specialised R&D departments in large corporations. They attempt to impose some analytical structure on these problems through three propositions concerning knowledge transfer and they argue that the 'make or buy' decisions of firms depend upon which kind of knowledge is required, providing a deeper understanding of this than is to be found in transaction cost economics.

Creativity results in new knowledge that is essential in any complex economic system. In his article, Jason Potts argues that creative service industries are laboratories for the generation of new ideas. This has been largely ignored in conventional approaches to understanding economic growth but Potts argues that this view fits very well with Schumpeter's vision. Correspondingly, he argues that the neo-Schumpeterian model should be revised to incorporate the origination, adoption, and retention of new rules emerging from the creative industries. It is noted that there has been much more emphasis upon science than the arts in arguments to develop an increased capability for innovation to occur. Potts argues that science and art are complementary and that neglect of the latter can constrain economic evolution. In this regard, he is somewhat critical of 'cultural economists' who have discussed the creative industries narrowly in terms of 'market failure' requiring government intervention through subsidies, tax breaks, etc. This leads to an understatement of their fundamental role in providing new knowledge in the evolutionary economic process. A number of examples are provided as to how creative industries shape economic evolution through the transmission of socio-cultural needs and aspirations, social technologies, and aesthetic innovations. Potts even speculates as to whether healthy creative industries are as fundamentally important for a healthy economic system as property rights and well-functioning

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free markets. As he reminds us, a recurring feature of oppressive, totalitarian regimes has been the restriction and control of creative industries.

In their article, Mario Cimoli and Gabriel Porcile also deal with knowledge but with a different emphasis, namely, the enhancement of technological capabilities through learning processes in developing economy contexts. The focus of attention is Latin America where there has been a strong macroeconomic-oriented 'structuralist' tradition in development economics for a number of decades. Cimoli and Porcile argue that it is fruitful to integrate this tradition with that of evolutionary economics to obtain a better understanding of the development process. Structuralist and related Keynesian considerations mean that learning and capability building must be accompanied by appropriate expansions in aggregate demand and Cimoli and Porcile examine, analytically, the conditions where an economy can make a knowledge traverse rather than being stuck in a hysteretic state. It is pointed out that debt crises in Latin American countries led to subsequent falls in aggregate demand with repercussions upon learning and capability building. They also note that the rise in commodity prices has pushed such economies away from diversified development towards traditional resource exports, again with repercussions for learning and innovation. Thus, a key role for the government in structurally changing developing economies is stressed, particularly in getting the institutional framework right and in encouraging a coherent national innovation system to develop.

The final two papers are devoted to more explicit approaches to evolutionary, complex system modelling. Complex systems are network structures and should be dealt with as networks, not collapsed into analytical functional relationships, such as the production functions that underpin most of conventional growth models. Paul Ormerod and Bridget Rosewell show how network simulations can enlighten us as to how cities, viewed as complex systems, grow and become more productive because of the formation of dense connectivity and associated spillovers. The agent-based approach that they employ belongs to a class of models that Thomas Schelling labelled as 'binary decisions with externalities'. Such modelling gives powerful support to some historical accounts of how cities grow and, the authors argue that such simulations could be calibrated over periods as short as a decade. It is clear that this kind of theorising through agent-based simulation offers a radically different way of modelling economic growth and is illustrative of the power of adopting a complex system perspective upon economic growth.

In the final article, Pietro Terna critically appraises the agent-based modelling methodology in its various manifestations. It is made clear that this methodological approach has weaknesses as well as strengths and that it remains very much a 'work in progress'. His main concern is to identify a particular approach which is appropriate to modelling innovation. He identifies two useful approaches to the construction of agent-based simulation models for this purpose and also attempts to deal with reinforcement learning. First, the Environment–Rules–Agents approach to structuring models is introduced and, second, it is shown how Cross Target schemes admit the possibility of using artificial neural networks to regulate agent behaviour in a realistic manner. Terna illustrates the approach by using the metaphor of Taormina's Chameleons, which can change their patterns of colour in a way that mimics epidemics spreading between connected agents in a network. A number of simulation experiments are conducted and a range of cases are related to actual historical cases of innovation diffusion. However, there is no attempt to reach a conclusion concerning the innovation process in this article. The main purpose is to introduce advanced students and researchers to a particular class of agent-based simulation models that employ a basic low-level language so that it is clear what is happening in the simulations. This has the advantage of making replication easier to undertake. However, it is the 'self-generating'

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character of rules in this methodological approach that makes it so attractive in the context of modelling innovation processes.

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